

Assignment:06

Q1. Create a abstract class Bank that has abstract method getROI(). Create two classes SBI, PNB, BOI inherited from Bank. Create a driver class that prints the rate of interest of each bank using super class memory reference.

```
Ans: abstract class Bank {  
  
    abstract double getROI();  
  
}  
  
class SBI extends Bank {  
  
    double getROI() {  
  
        return 6.5;  
  
    }  
  
}  
  
class PNB extends Bank {  
  
    double getROI() {  
  
        return 7.0;  
  
    }  
  
}  
  
class BOI extends Bank {  
  
    double getROI() {  
  
        return 6.8;  
  
    }  
  
}  
  
public class Main {    public static void  
main(String[] args) {    Bank b;    b  
= new SBI();  
  
    System.out.println("SBI Rate of  
Interest: " + b.getROI() + "%");
```

```

        b = new PNB();

        System.out.println("PNB Rate of Interest: " + b.getROI() + "%");

    b = new BOI();

        System.out.println("BOI Rate of Interest: " + b.getROI() + "%");

    }

}

```

output:

```

D:\SAINA CHOUDHURY\25BCSL32>javac Main.java

D:\SAINA CHOUDHURY\25BCSL32>java Main
SBI Rate of Interest: 6.5%
PNB Rate of Interest: 7.0%
BOI Rate of Interest: 6.8%

D:\SAINA CHOUDHURY\25BCSL32>|

```

Q2. Define an interface Calculator which has the basic methods add(), sub(), mul() and div(). Define a concrete class named as DemoCalculator that implements the interface. Define the driver class, which create object reference of the interface Calculator and perform all basic operation of the calculator.

Ans: interface Calculator {

```

double add(double a, double b);

double sub(double a, double b);

double mul(double a, double b);

double div(double a, double b);

}

class DemoCalculator implements Calculator {

public double add(double a, double b) {

    return a + b;

}

public double sub(double a, double b) {

    return a – b;

}

}

```

```

    } public double mul(double a, double b) {
        return a * b;
    }

    public double div(double a, double b) {
        if (b == 0) {
            System.out.println("Division by zero is not allowed!");
            return 0;
        }
        return a / b;
    }
}

public class Main {    public static void
main(String[] args) {    Calculator calc =
new DemoCalculator();    double a = 20, b
= 5;

    System.out.println("Addition: " + calc.add(a, b));
    System.out.println("Subtraction: " + calc.sub(a, b));
    System.out.println("Multiplication: " + calc.mul(a, b));
    System.out.println("Division: " + calc.div(a, b));
}
}

```

Output:

```

D:\SAINA CHOUDHURY\25BCSL32>javac Driver.java
D:\SAINA CHOUDHURY\25BCSL32>java Driver
Addition: 25.0
Subtraction: 15.0
Multiplication: 100.0
Division: 4.0
D:\SAINA CHOUDHURY\25BCSL32>

```

Q3. Create an abstract class 'Shape' with three abstract methods namely 'RectangleArea' taking two parameters, 'SquareArea' and 'CircleArea' taking one parameter each. The

parameters of 'RectangleArea' are its length and breadth, that of 'SquareArea' is its side and that of 'CircleArea' is its radius. Now create another class 'Area' containing all the three methods 'RectangleArea', 'SquareArea' and 'CircleArea' for printing the area of rectangle, square and circle respectively. Create an object of class 'Area' and call all the three methods.

```
Ans: abstract class Shape {    abstract void
RectangleArea(double length, double breadth);    abstract void
SquareArea(double side);    abstract void CircleArea(double
radius);
}
class Area extends Shape {    void
RectangleArea(double length, double breadth) {
double area = length * breadth;
    System.out.println("Area of Rectangle: " + area);
}
    void SquareArea(double side) {
double area = side * side;
    System.out.println("Area of Square: " + area);
}
    void CircleArea(double radius) {
double area = Math.PI * radius * radius;
    System.out.println("Area of Circle: " + area);
}
} public class Main {    public static void
main(String[] args) {        Area obj =
new Area();        obj.RectangleArea(10,
5);        obj.SquareArea(4);
obj.CircleArea(3);
}
```

```
}
```

Output:

```
D:\SAINA CHOUDHURY\25BCSL32>javac Area_Main.java
D:\SAINA CHOUDHURY\25BCSL32>java Area_Main
Area of Rectangle: 50.0
Area of Square: 16.0
Area of Circle: 28.274333882308138
D:\SAINA CHOUDHURY\25BCSL32>|
```

Q4. Write a program to implement multiple inheritance using interface.

Ans:

```
interface Printable {
    void print();
}

interface Showable {
    void show();
}

class Demo implements Printable, Showable {
    public void print() {
        System.out.println("This is the print() method.");
    }
    public void show() {
        System.out.println("This is the show() method.");
    }
}

public class Main {
    public static void
    main(String[] args) {
        Demo obj =
        new Demo();

        obj.print();

        obj.show();
    }
}
```

```
}  
}
```

Output:

```
D:\SAINA CHOUDHURY\25BCSL32>javac Multiple_Inheritance.java  
D:\SAINA CHOUDHURY\25BCSL32>java Multiple_Inheritance  
This is the print() method.  
This is the show() method.  
D:\SAINA CHOUDHURY\25BCSL32>|
```

Q5. Create an interface Servicing that has abstract methods getServiceTime(). Create two class Car, Bike that implement interface. Create a driver class that creates the objects of two class and displays the service time.

Ans:

```
interface Servicing {  
    void getServiceTime();  
}  
  
class Car implements Servicing {  
    public void getServiceTime() {  
        System.out.println("Car service time: Every 6 months or 5000 km.");  
    }  
}  
  
class Bike implements Servicing {  
    public void getServiceTime() {  
        System.out.println("Bike service time: Every 3 months or 2000 km.");  
    }  
}  
  
public class Main {    public static void  
main(String[] args) {  
    Servicing car = new Car();  
  
    Servicing bike = new Bike();  
  
    car.getServiceTime();  
  
    bike.getServiceTime();  
}
```

}

Output:

```
D:/2VIMV CHONDHANBA/S2BC2G35>
BTK6 26LA7C6 F7W6: EΛ6LΛ 3 WOUFμ2 OL 3000 Kμ'.
C9L 26LA7C6 F7W6: EΛ6LΛ 6 WOUFμ2 OL 2000 Kμ'.
D:/2VIMV CHONDHANBA/S2BC2G35>]9Λ9 DL7Λ6L~26LA7C7U0~]9Λ9
D:/2VIMV CHONDHANBA/S2BC2G35>]9Λ9C DL7Λ6L~26LA7C7U0~]9Λ9
```

Q6. Create a Package **btech** which has one class **Student**. Accept student detail through parameterized constructor of **Student** class. Write a method **display()** to display the student details. Create another class **Test** containing the main method which will use the package **btech** and calculate total marks and percentage of marks. One sample output is shown below.

```
D:\>javac -d . Student.java
D:\>javac StudentMain.java
D:\>java StudentMain
Enter Roll no:= 101
Enter Name:= Abhay
Enter 3 sub mark:= 87 56 91
Roll_no : 101
Name    : Abhay
-----MARKS-----
Sub 1   : 87
Sub 2   : 56
Sub 3   : 91
Total   : 234
percentage: 78
-----
```

Ans:// student file

```
package btech; public
```

```
class Student {
```

```
    String name;
```

```
    int rollNo; int m1, m2, m3; public Student(String name, int
```

```
rollNo, int m1, int m2, int m3) {    this.name = name;
```

```
this.rollNo = rollNo;    this.m1 = m1;    this.m2 = m2;
```

```
this.m3 = m3;
```

```
}
```

```
public void display() {
```

```
    System.out.println("Student Name: " + name);
```

```
    System.out.println("Roll No: " + rollNo);
```

```
    System.out.println("Marks in Subject 1: " + m1);
```

```
    System.out.println("Marks in Subject 2: " + m2);
```

```
    System.out.println("Marks in Subject 3: " + m3);
```



```

    }

    public int totalMarks() {
return m1 + m2 + m3;
    }

    public double percentage() {
return (totalMarks() / 3.0);
    }
}

// test file
import btech.Student; import
java.util.Scanner; public class
Test {

    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter Student Name: ");
        String name = sc.nextLine();
        System.out.print("Enter Roll Number: ");
int roll = sc.nextInt();

        System.out.print("Enter marks in 3 subjects: ");
int m1 = sc.nextInt();    int m2 = sc.nextInt();
int m3 = sc.nextInt();

        Student s = new Student(name, roll, m1, m2, m3);
System.out.println("\n--- Student Details ---");

        s.display();

        System.out.println("Total Marks: " + s.totalMarks());
System.out.println("Percentage: " + s.percentage() + "%");

        sc.close();
    }
}

```

```
}  
}
```

Output:

```
C:\Users\Public\Saina Choudhry\25bcs135\Assignment_6>java te  
st  
C:\Users\Public\Saina Choudhry\25bcs135\Assignment_6>java te  
st  
Enter Student Name: Saina  
Enter Roll Number: 34  
Enter marks in 3 subjects: 89 98 95  
  
--- Student Details ---  
Student Name: Saina  
Roll No: 34  
Marks in Subject 1: 89  
Marks in Subject 2: 98  
Marks in Subject 3: 95  
Total Marks: 282  
Percentage: 94.0%
```

Q7. Create a sub-package called arithmetic under the package btech. The arithmetic package should contain a class MyMath having methods to deal with different arithmetic operations (addition, subtraction, multiplication, division and mod). Create a class Test containing the main method which will use the methods of sub-package arithmetic.

Ans: package

btech.arithmetic; public

```
class MyMath {  
    public int add(int a, int b) {  
        return a + b;  
    }  
    public int subtract(int a, int b) {  
        return a - b;  
    }  
    public int multiply(int a, int b) {  
        return a * b;  
    }  
    public double divide(int a, int b) {  
        if (b == 0) {  
            System.out.println("Division by zero is not allowed!");  
        }  
    }  
}
```

```
        return 0;
    }
    return (double) a / b;
}

public int mod(int a, int b) {
return a % b;
}
}

//main java file import
btech.arithmetic.MyMath; import
java.util.Scanner; public class Test
{   public static void
main(String[] args) {
    Scanner sc = new Scanner(System.in);
    MyMath math = new MyMath();
    System.out.print("Enter first number: ");
int num1 = sc.nextInt();
    System.out.print("Enter second number: ");
int num2 = sc.nextInt();

    System.out.println("\n--- Arithmetic Operations ---");
    System.out.println("Addition: " + math.add(num1, num2));
    System.out.println("Subtraction: " + math.subtract(num1, num2));
    System.out.println("Multiplication: " + math.multiply(num1, num2));
    System.out.println("Division: " + math.divide(num1, num2));
    System.out.println("Modulus: " + math.mod(num1, num2));
    sc.close();
}
```

```
}
```

Output:

```
C:\Users\Public\Saina Choudhry\25bcs135\Assignment_6
javac Test.java
C:\Users\Public\Saina Choudhry\25bcs135\Assignment_6
Enter first number: 25
Enter second number: 30

--- ARITHMETI OPERATIONS ---
Addition: 55
Subtraction: -5
Multiplication: 750
Division: 0.8333333333333334
Modulus: 25
```

Q8. Create a sub-package named **shapes** under a package **org**. Create some classes in the package representing some common geometric shapes like **Square**, **Triangle**, **Circle** and so on. The classes should contain the **area()** and **perimeter()** methods in them. Compile the package. Use this package to find area and perimeter of different shapes as chosen by the user.

Ans:// square file

```
package org.shapes; public
class Square {    double side;
public Square(double side) {
    this.side = side;
    }
    public double area() {
return side * side;
    }
    public double perimeter() {
return 4 * side;
    }
}
```

// triangle file

```

package org.shapes; public class Triangle {
double a, b, c;  public Triangle(double a,
double b, double c) {
    this.a = a;
this.b = b;    this.c
= c;
    }
    public double perimeter() {
return a + b + c;
    }
    public double area() {
        double s = perimeter() / 2;
        return Math.sqrt(s * (s - a)
        * (s - b) * (s - c));
    }
}

```

// circle file

```

package org.shapes; public
class Circle {  double radius;
public Circle(double radius) {
this.radius = radius;
    }
    public double area() {    return
Math.PI * radius * radius;
    }
    public double perimeter() {
return 2 * Math.PI * radius;

```

```

    }
}

// TestShape file import org.shapes.*;
import java.util.Scanner; public class
TestShapes {    public static void
main(String[] args) {

    Scanner sc = new Scanner(System.in);

    System.out.println("--- Choose Shape ---");

    System.out.println("1. Square");

    System.out.println("2. Triangle");

    System.out.println("3. Circle"); System.out.print("Enter your choice: ");    int choice =
    sc.nextInt();    switch (choice) {        case 1:

        System.out.print("Enter side of square: ");

double s = sc.nextDouble();

        Square square = new Square(s);

        System.out.println("Area of Square: " + square.area());

        System.out.println("Perimeter of Square: " + square.perimeter());

        break;

    case 2:

        System.out.print("Enter three sides of triangle: ");

double a = sc.nextDouble();        double b =
sc.nextDouble();        double c = sc.nextDouble();

        Triangle triangle = new Triangle(a, b, c);

        System.out.println("Area of Triangle: " + triangle.area());

        System.out.println("Perimeter of Triangle: " + triangle.perimeter());

        break;

    case 3:

```

```

        System.out.print("Enter radius of circle: ");
double r = sc.nextDouble();          Circle circle =
new Circle(r);

        System.out.println("Area of Circle: " + circle.area());

        System.out.println("Perimeter of Circle: " + circle.perimeter());

        break;

default:

        System.out.println("Invalid choice!");

    }

}

}

```

Output:

```

C:\Users\Public\Saina Choudhry\25bcs135\Assignment_6
javac TestShapes.java
C:\Users\Public\Saina Choudhry\25bcs135\Assignment_6
---- CHOOSE SHAPE ---
1. Square
2. Triangle
3. Circle
Enter your choice: 3
Enter radius of circle: 3.55
Area of Circle: 39.5912914168636
Perimeter of Circle: 22.305307804048753

```